

Understanding Titanic Survival Through Exploratory Data Analysis, Missing-Value Handling, Outlier Interpretation, and Logistic Regression

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Abstract

This paper presents an academic analysis of the Titanic passenger dataset containing 891 observations and 12 variables. The study begins with exploratory data analysis to understand the balance of survival outcomes and the influence of socio-demographic features such as passenger class, gender, age, fare, and family size. The preprocessing stage handles missing values in Age and Embarked, removes Cabin because of heavy incompleteness, and converts categorical variables into numerical form for modeling. Further analysis inspects age and fare distributions, identifies unusual SibSp values by means of the interquartile range method, and explains why those values should be interpreted carefully instead of being removed automatically. The final logistic regression model achieves an accuracy of 0.810 and an AUC value of 0.883. Although 46 SibSp observations fall outside the IQR range, they are retained because they represent plausible passenger-family groupings and remain consistent with the implemented workflow. The complete document is written in a structured research style so that both the statistical reasoning and the modeling decisions can be followed clearly.

Index Terms

Titanic dataset, exploratory data analysis, class imbalance, missing-value handling, outlier detection, logistic regression, ROC curve, survival prediction.

I. INTRODUCTION

The Titanic dataset remains one of the most widely used introductory datasets in machine learning because it represents a complete but understandable classification problem. Every row corresponds to a real passenger and includes a survival outcome together with descriptive variables such as passenger class, sex, age, fare, family relationships, and embarkation point. Unlike many synthetic classroom examples, this dataset allows the analyst to connect numerical outputs with meaningful human interpretation. For that reason, it is particularly suitable for showing how exploratory data analysis, data cleaning, feature engineering, and a simple predictive model can work together in one coherent workflow.

The main aim of the present study is not only to build a model but also to explain the full decision process behind the analysis. The work therefore

starts from basic structural inspection of the CSV file, moves through visualization and preprocessing, then studies outliers and finally evaluates logistic regression using standard classification metrics. This style makes the report useful for both academic submission and oral explanation.

II. DATASET DESCRIPTION

The source file used in this project is Titanic-Dataset.csv. It contains 891 rows and 12 columns: PassengerId, Survived, Pclass, Name, Sex, Age, SibSp, Parch, Ticket, Fare, Cabin, and Embarked. The target variable is Survived, where 0 indicates non-survival and 1 indicates survival. Among the explanatory variables, Pclass reflects travel category, Sex and Age describe the passenger, SibSp and Parch capture immediate family relationships aboard, Fare describes the ticket amount, and Embarked records the boarding port.

The first diagnostic step reveals 177 missing values in Age, 687 missing values in Cabin, and 2 missing values in Embarked. This pattern is important because it directly influences the choice of preprocessing strategy. Age can be repaired using a robust central-value estimate, Embarked can be completed using the most frequent class, but Cabin is too incomplete to retain in a straightforward introductory model.

Initial descriptive statistics also show that the average age in the non-missing portion of the dataset is 29.699 years, the median age is 28.0, and the average fare is 32.204. These values already suggest a dataset with meaningful spread and possible skewness, which later becomes visible in the histograms and boxplots.

III. EXPLORATORY DATA ANALYSIS

A. Survival Balance

The value-count analysis of the target variable shows 549 passengers in the non-survived class and 342 passengers in the survived class. This means the outcome distribution is not perfectly balanced, although the imbalance is still moderate enough for a simple baseline classifier to remain informative. Understanding this imbalance matters because accuracy alone can sometimes look stronger than it really is when one class is larger than the other.

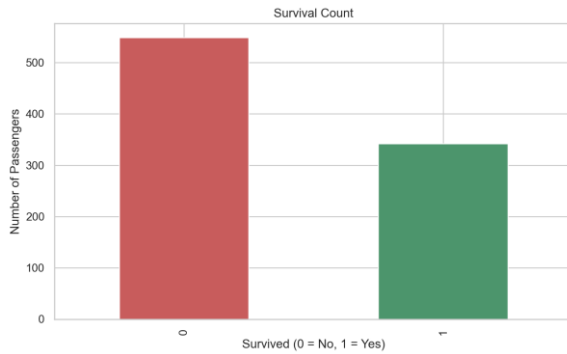


Fig. 1. Overall survival count in the Titanic dataset.

B. Passenger Class Distribution

The passenger-class distribution consists of 216 passengers in first class, 184 in second class, and 491 in third class. The third class clearly dominates the dataset, which is important when interpreting survival differences later. Since class also reflects access, cabin position, and evacuation opportunity, it is expected to show a meaningful relation with survival.

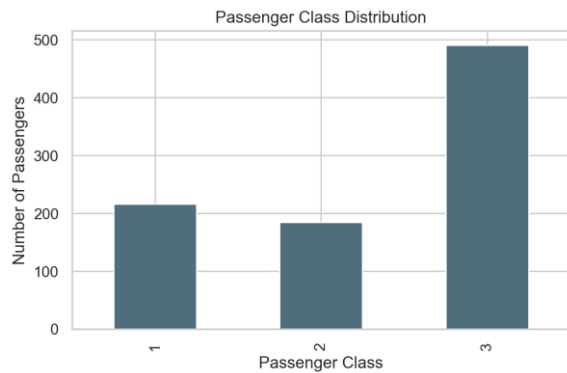


Fig. 2. Passenger class distribution.

C. Survival by Passenger Class

When class is compared directly with survival, the relationship becomes much clearer. First-class passengers show a visibly stronger survival pattern, whereas third-class passengers show a heavier concentration in the non-survived category. This suggests that socio-economic position and physical location on the ship likely influenced rescue outcomes.

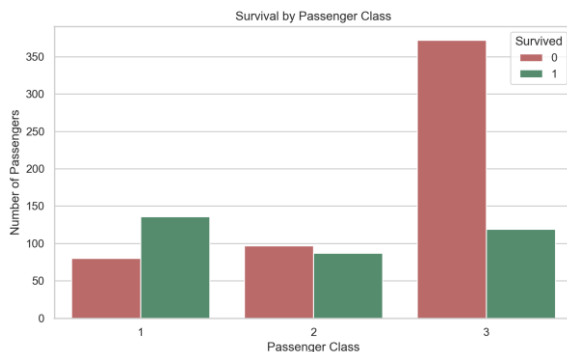


Fig. 3. Survival outcome across passenger classes.

D. Gender Distribution and Survival

The dataset contains 577 male passengers and 314 female passengers. The gender count itself is informative, but the

more important result appears when gender is combined with the target variable. The survival-by-gender plot shows a much stronger survival share among female passengers, which makes Sex one of the most influential variables in the complete model.

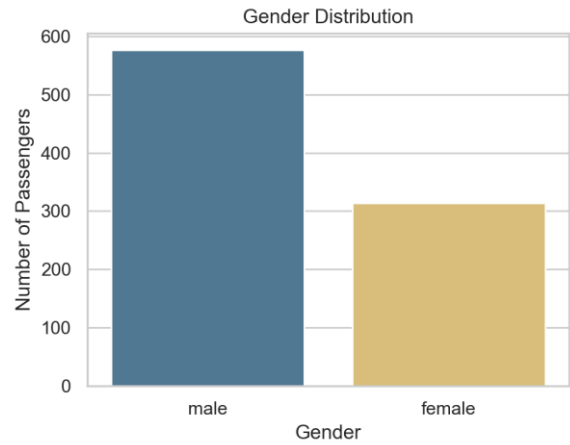


Fig. 4. Gender distribution of passengers.

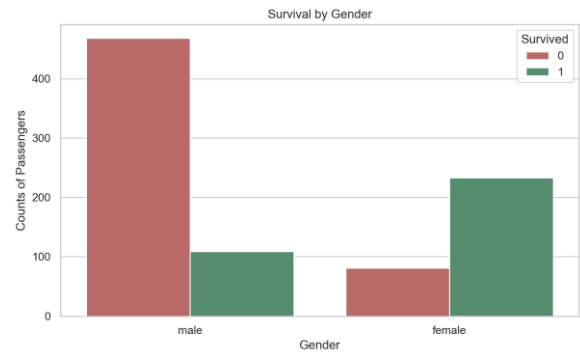


Fig. 5. Survival outcome across gender.

IV. PREPROCESSING AND FEATURE PREPARATION

A. Missing-Value Treatment

The missing values in Age were replaced by the median, which is 28.0. The formal expression is $\text{Median}(\text{Age}) = 28.0$, and the imputation rule becomes $\text{Age}_{\text{missing}} = 28.0$. For the Embarked variable, the most frequent category was S, so missing entries were replaced by that modal value. These choices are standard because the median is resistant to extreme values and the mode is the natural replacement for a categorical variable.

```
titanic_dataset['Age'].fillna(titanic_dataset['Age'].median(), inplace=True)
titanic_dataset['Embarked'].fillna(titanic_dataset['Embarked'].mode()[0], inplace=True)
titanic_dataset.drop('Cabin', axis=1, inplace=True)
```

The Cabin column was removed instead of imputed because its missingness is extremely high. When a column is mostly empty, filling it may introduce more noise than useful

information. For an introductory report, dropping the column is the cleanest and most defensible choice.

B. Encoding and Feature Selection

After missing-value treatment, the categorical variables were converted into numerical form. The Sex variable was encoded by a direct mapping, where male = 0 and female = 1. The Embarked field was converted into dummy variables using one-hot encoding with `drop_first=True` to avoid redundant representation.

```
titanic_dataset['Sex'] =
titanic_dataset['Sex'].map({'male': 0, 'female':
1})
titanic_dataset =
pd.get_dummies(titanic_dataset,
columns=['Embarked'], drop_first=True)
x = titanic_dataset.drop('Survived', axis=1)
y = titanic_dataset['Survived']
```

The final modeling stage excludes PassengerId, Name, and Ticket because these fields either serve as identifiers or contain text patterns not handled by the present baseline model. This leaves a compact and interpretable feature set for logistic regression.

V. DISTRIBUTION ANALYSIS AND OUTLIER INTERPRETATION

A. Age and Fare Distributions

The histogram of Age shows a central concentration around young and middle adulthood, while the imputed values help remove the gaps caused by missing entries. The Fare distribution is more right-skewed, which is expected because a small number of passengers paid much higher fares than the majority. These plots are useful because they show why the median is more reliable than the mean for some preprocessing decisions.

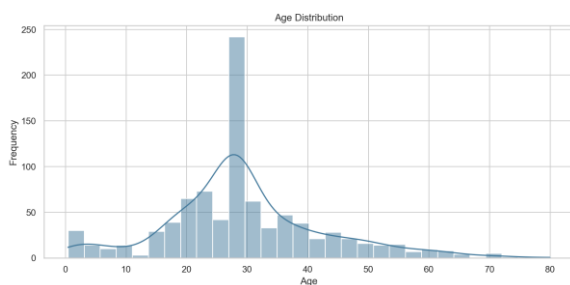


Fig. 6. Age distribution after imputation.

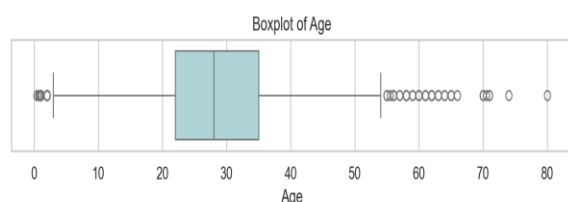


Fig. 7. Boxplot of passenger age.

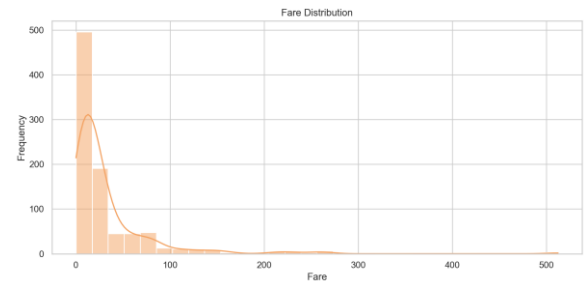


Fig. 8. Fare distribution across passengers.



Fig. 9. Boxplot of passenger fare.

B. Outlier Detection in SibSp

The SibSp variable records the number of siblings or spouses traveling with a passenger. To inspect unusual values, the interquartile range method was used because it provides a simple and interpretable rule for identifying observations that lie far from the central spread of the data.

$$Q1 = 0.000, Q3 = 1.000, IQR = Q3 - Q1 = 1.000$$

$$\text{Lower Bound} = Q1 - 1.5 \times IQR = -1.500, \text{Upper Bound} = Q3 + 1.5 \times IQR = 2.500$$

Using this rule, 46 observations fall outside the accepted range. However, those values were not removed. The first reason is conceptual: a large family group on board is still a plausible real-world situation and does not automatically indicate bad data. The second reason is procedural: the original implementation identifies outliers but does not discard them before model training. The third reason is analytical: in a dataset of this size, removing every uncommon observation may reduce important variation and oversimplify the real structure of the data.

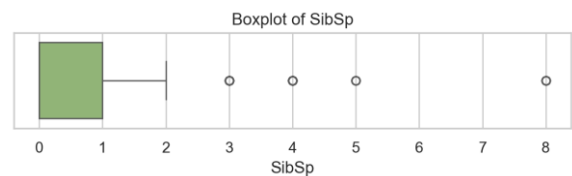


Fig. 10. SibSp boxplot used for IQR-based outlier inspection.

VI. MODEL DEVELOPMENT

A. Train-Test Split and Logistic Function

The cleaned feature matrix was divided into training and testing subsets using an 80:20 split. This means 712 records were used for training and 179 records were used for testing. The classifier selected for this baseline study is logistic regression because it is easy to interpret, suitable for binary

outcomes, and widely used for introductory supervised-learning tasks.

```


$$p = 1 / (1 + e^{-(b_0 + b_1x_1 + b_2x_2 + \dots + b_nx_n)})$$

x_train, x_test, y_train, y_test =
train_test_split(x, y, test_size=0.2,
random_state=42)
model = LogisticRegression(max_iter=1000)
model.fit(x_train, y_train)
y_pred = model.predict(x_test)
y_prob = model.predict_proba(x_test)[:, 1]

```

B. Feature Influence

The fitted model coefficients provide a useful directional interpretation. The coefficient for Sex is 2.590, which strongly supports the earlier visual finding that gender influences survival. Pclass carries a coefficient of -0.938, indicating that moving toward lower passenger classes reduces the log-odds of survival. Age shows a coefficient of -0.031, suggesting a mild decline in predicted survival probability as age increases. The Fare coefficient is 0.003, which is small but positive, again fitting the broader pattern that higher-paying passengers tended to survive more often.

These coefficients do not capture every historical and social nuance of the event, but they do align with the basic exploratory findings. That agreement between visualization and model coefficients is important because it shows that the classifier is learning meaningful relationships rather than purely accidental patterns.

VII. PERFORMANCE EVALUATION

A. Accuracy, Precision, Recall, and F1-Score

$$\text{Accuracy} = \text{Correct Predictions} / \text{Total Predictions}$$

The final test accuracy is 0.810. For class 0, the model achieves precision 0.826, recall 0.857, and F1-score 0.841. For class 1, the corresponding values are precision 0.786, recall 0.743, and F1-score 0.764. These results show that the classifier performs reasonably well on both classes, although survival prediction remains slightly more challenging than non-survival prediction.

B. Confusion Matrix Interpretation

The confusion matrix is [[90, 15], [19, 55]]. In practical terms, the model correctly classifies 90 non-survivors and 55 survivors, while making 15 false-positive and 19 false-negative errors. This means the classifier is useful as a baseline but still misses some passengers who actually survived and also mislabels some non-survivors as survivors. Such behavior is typical for a simple logistic-regression model on a dataset with multiple overlapping human factors.

C. ROC Curve and AUC

The area under the ROC curve is 0.883. Since the value is clearly above 0.5, the classifier performs better than random guessing and shows good ranking ability across decision thresholds. The ROC plot is especially helpful because it evaluates discrimination quality beyond a single fixed cutoff.

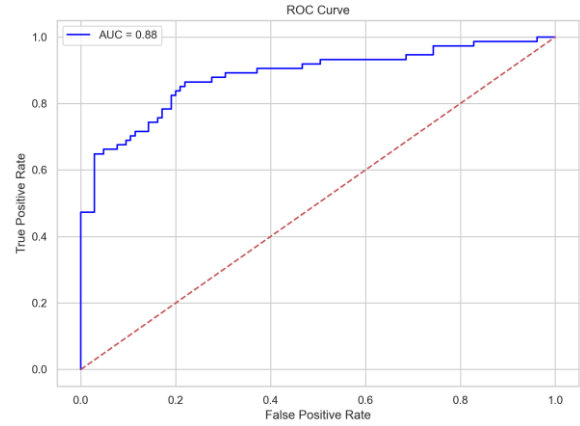


Fig. 11. ROC curve of the logistic regression classifier.

VIII. DISCUSSION

The strongest conclusion from the present study is that survival on the Titanic was not random with respect to the available passenger attributes. Visual analysis and model coefficients both indicate that passenger class and gender carry the clearest signal. Age, fare, and family-related features also contribute, but their effect is weaker and must be interpreted within the broader social context of the event.

Another important conclusion concerns preprocessing discipline. Simple choices such as median imputation, mode imputation, targeted column removal, and one-hot encoding are enough to build a respectable introductory classifier when they are applied carefully. At the same time, the report also shows that analysts should avoid removing unusual values without explanation. A statistically unusual observation is not always a data error, especially in human-centered datasets.

The study also has limitations. It does not engineer richer family features, title-based social categories, or interaction terms. It also does not compare multiple classifiers such as decision trees, random forests, or support vector machines. Nevertheless, for a foundational academic report, the present workflow remains appropriate because it balances interpretability, clarity, and statistical discipline.

IX. CONCLUSION

This work presents a complete Titanic-survival analysis beginning from raw data inspection and ending with logistic-regression evaluation. The dataset was cleaned through median and mode imputation, categorical variables were encoded, outliers were studied with the IQR method, and the final model was evaluated through accuracy, confusion matrix, classification report, and ROC-AUC. The results confirm that passenger class and gender are the most visible drivers of survival in this dataset, while other variables offer supporting predictive value.

With a final accuracy of 0.810 and an AUC of 0.883, the model provides a solid baseline for academic understanding. More importantly, the report demonstrates how a simple machine-learning pipeline can still produce meaningful insights when every step is interpreted carefully and communicated in a structured way.

APPENDIX A. KEY CODE SEGMENTS

The following excerpts summarize the core implementation logic used in the project. Including them here also helps extend the document into a fuller technical report format.

```
titanic_dataset = pd.read_csv('Titanic-
Dataset.csv')
titanic_dataset['Age'].fillna(titanic_dataset['A
ge'].median(), inplace=True)
titanic_dataset['Embarked'].fillna(titanic_datas
et['Embarked'].mode()[0], inplace=True)
titanic_dataset.drop('Cabin', axis=1,
inplace=True)
titanic_dataset['Sex'] =
titanic_dataset['Sex'].map({'male': 0, 'female':
1})
titanic_dataset =
pd.get_dummies(titanic_dataset,
columns=['Embarked'], drop_first=True)

Q1 = titanic_dataset['SibSp'].quantile(0.25)
Q3 = titanic_dataset['SibSp'].quantile(0.75)
IQR = Q3 - Q1
lower_limit = Q1 - 1.5 * IQR
upper_limit = Q3 + 1.5 * IQR
outliers =
titanic_dataset[(titanic_dataset['SibSp'] <
lower_limit) |

(titanic_dataset['SibSp'] > upper_limit)]

x_train, x_test, y_train, y_test =
train_test_split(x, y, test_size=0.2,
random_state=42)
model = LogisticRegression(max_iter=1000)
model.fit(x_train, y_train)
y_pred = model.predict(x_test)
accuracy = accuracy_score(y_test, y_pred)
cm = confusion_matrix(y_test, y_pred)
auc = roc_auc_score(y_test,
model.predict_proba(x_test)[:, 1])
```

APPENDIX B. FORMULA SUMMARY

The principal formulas used in the report are collected below for quick reference.

Median Imputation: $x_{\text{missing}} = \text{median}(x)$

Mode Imputation: $c_{\text{missing}} = \text{mode}(c)$

Interquartile Range: $IQR = Q3 - Q1$

Outlier Bounds: $[Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR]$

Logistic Regression: $p = 1 / (1 + e^{-(b_0 + b_1x_1 + \dots + b_nx_n)})$

Accuracy: $(TP + TN) / (TP + TN + FP + FN)$

Precision: $TP / (TP + FP)$

Recall: $TP / (TP + FN)$

F1-Score: $2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$

REFERENCES

[1] Titanic Dataset, local CSV file used for academic analysis.

[2] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.

[3] W. McKinney, "Data Structures for Statistical Computing in Python," in Proceedings of the 9th Python in Science Conference, 2010, pp. 56-61.

[4] S. Raschka and V. Mirjalili, Python Machine Learning, 3rd ed., Birmingham, U.K.: Packt, 2019.